EBN 122

Chapter 5:
Operational Amplifiers
(Op Amps)

Chapter 5 overview

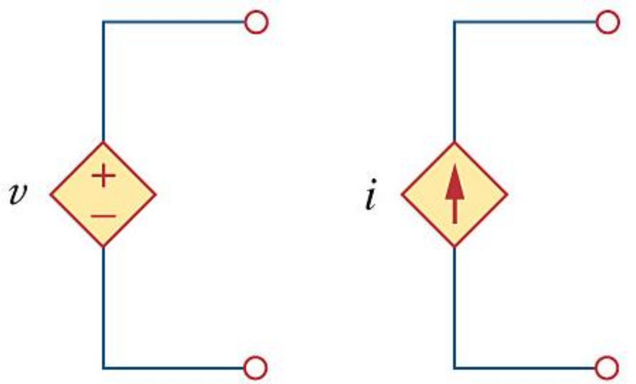
5.2. Operational Amplifiers (Op Amps)
5.3. Ideal Op Amps

Video lecture 1

5.4. Inverting Amplifiers
5.5. Noninverting Amplifiers
5.6. Summing Amplifiers
5.7. Difference Amplifiers
5.8. Cascaded Amplifier Circuits

What is an Op Amp?

- Electronic unit that behaves like a **voltage-controlled voltage source**



- Active circuit element

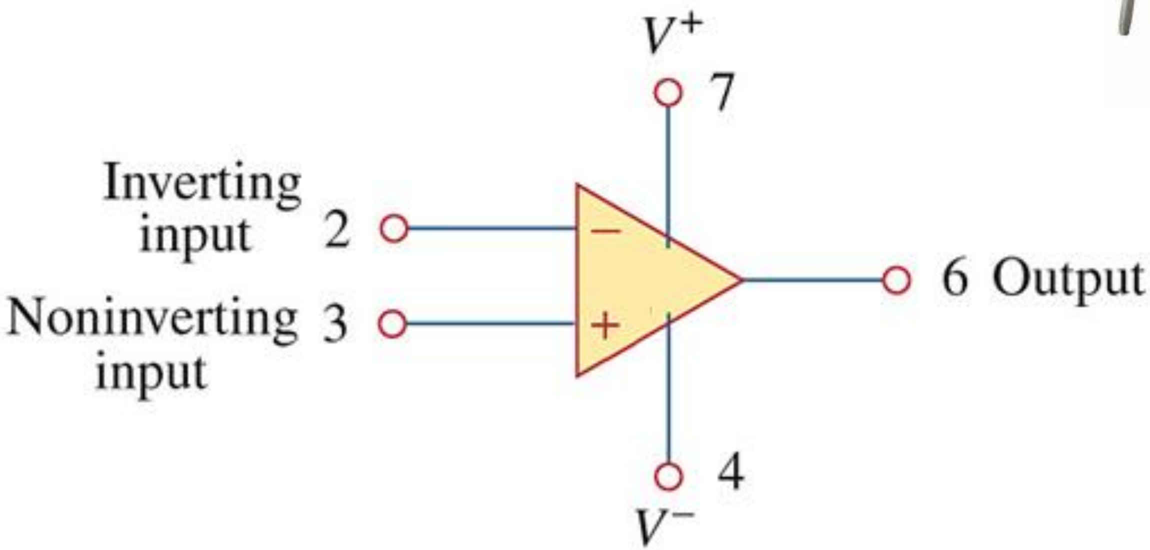
- Designed to perform **mathematical <u>operations:</u>**

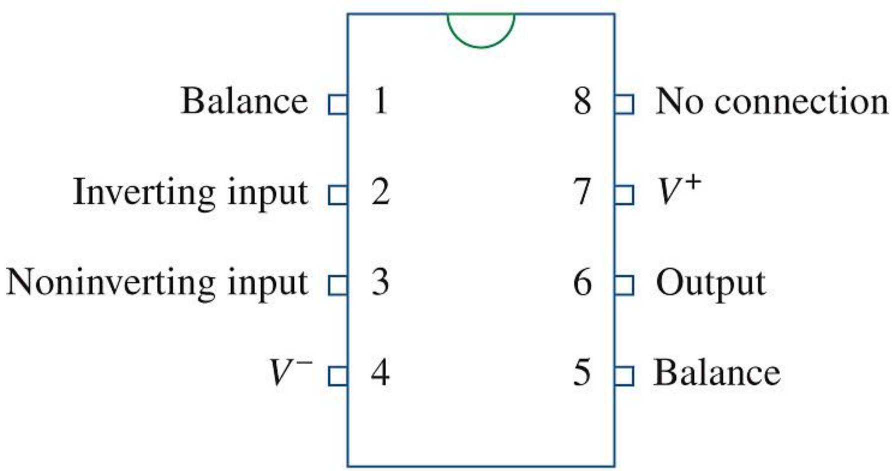
→ addition, subtraction, multiplication, division, differentiation, integration

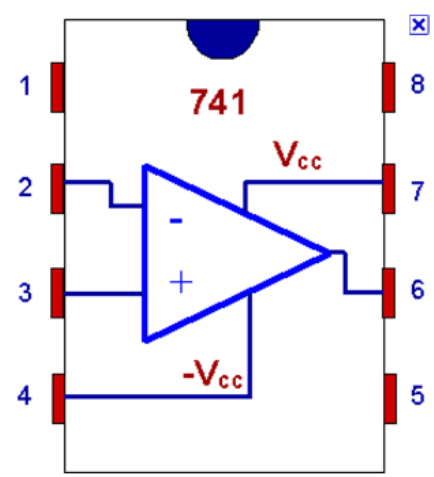
- Consists of complex arrangement of components:

→ resistors, transistors, capacitors, diodes

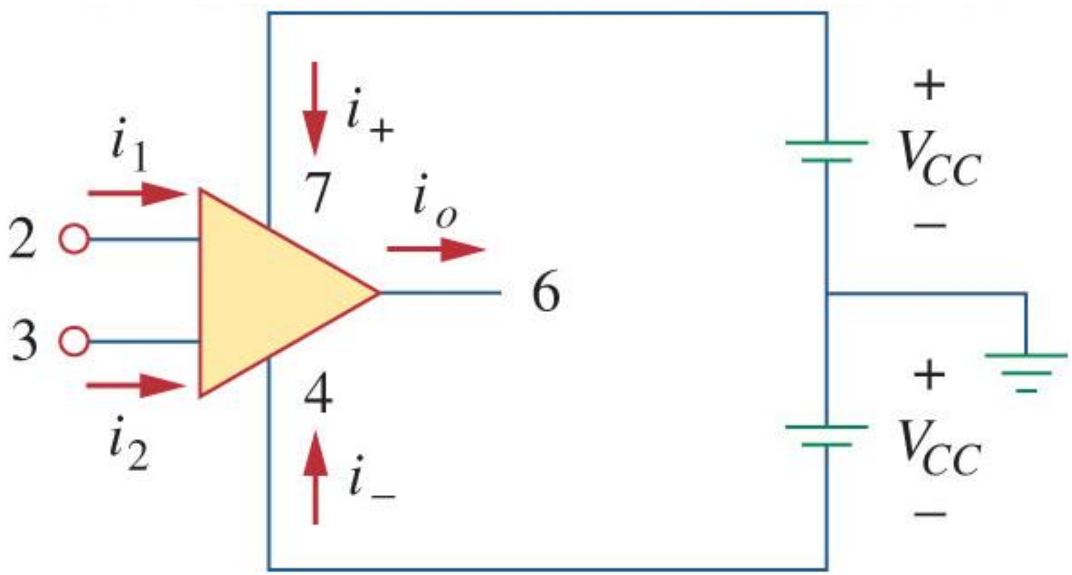
What is an Op Amp?







What is an Op Amp?

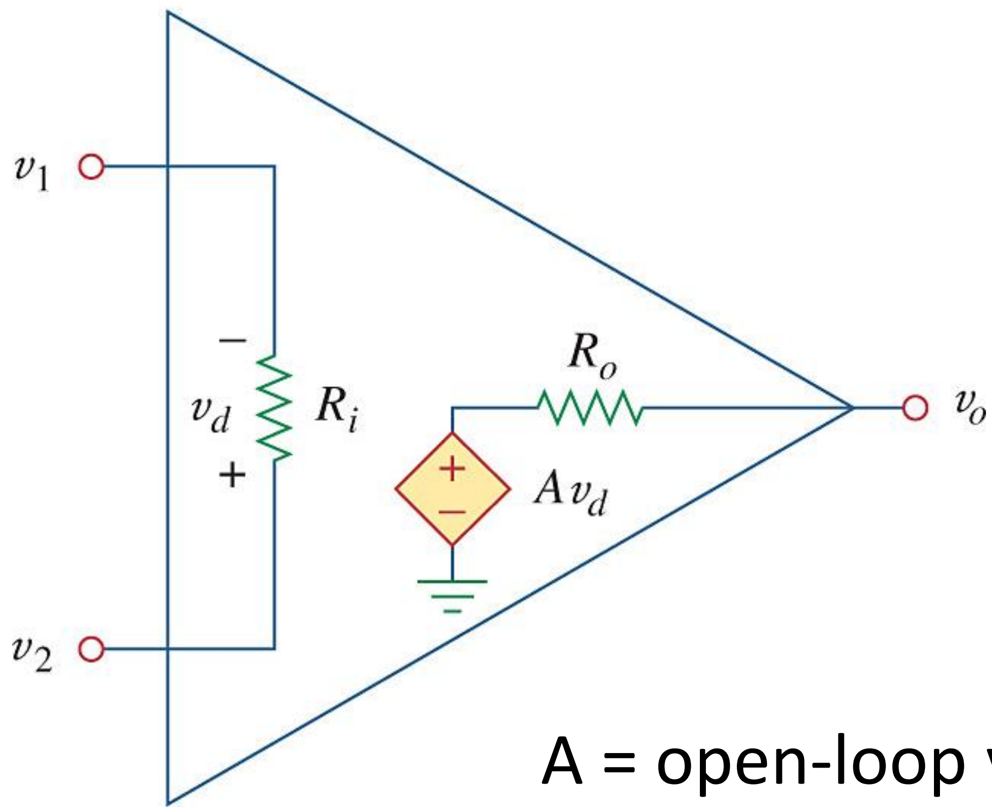


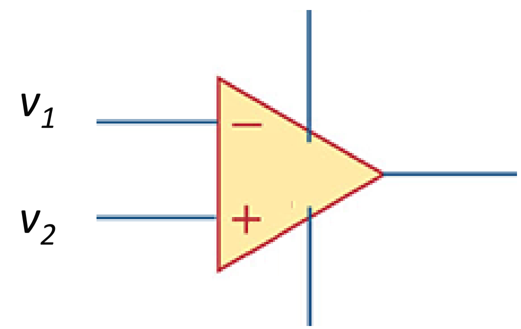
KCL:

$i_0 = i_1 + i_2 + i_+ + i_-$

Maximum and minimum limits for output voltage of op-amp (v_o) as a result of power supply voltages

Equivalent circuit for op-amp





$$v_d = v_2 - v_1$$

$$v_o = Av_d = A(v_2 - v_1)$$

A = open-loop voltage gain (no external feedback from output v_o to input v_d)

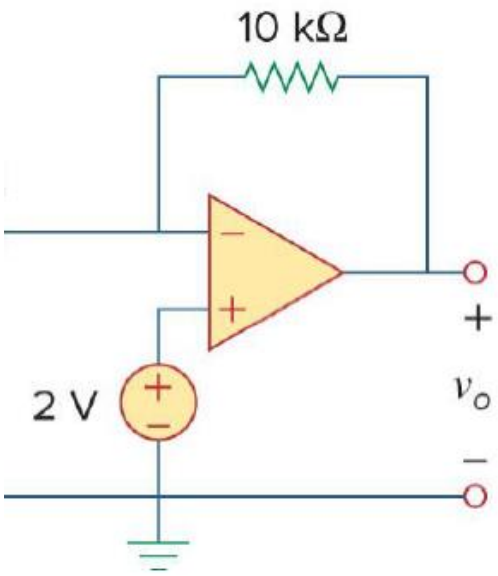
Op-amp parameters and properties

Parameter	Range	Ideal
A	$10^5 - 10^8$	∞
R_i	$10^5 - 10^{13}\,\Omega$	$\infty\,\Omega$
R_o	$10 - 100\,\Omega$	$0\,\Omega$
V_{CC}	5 – 24 V	

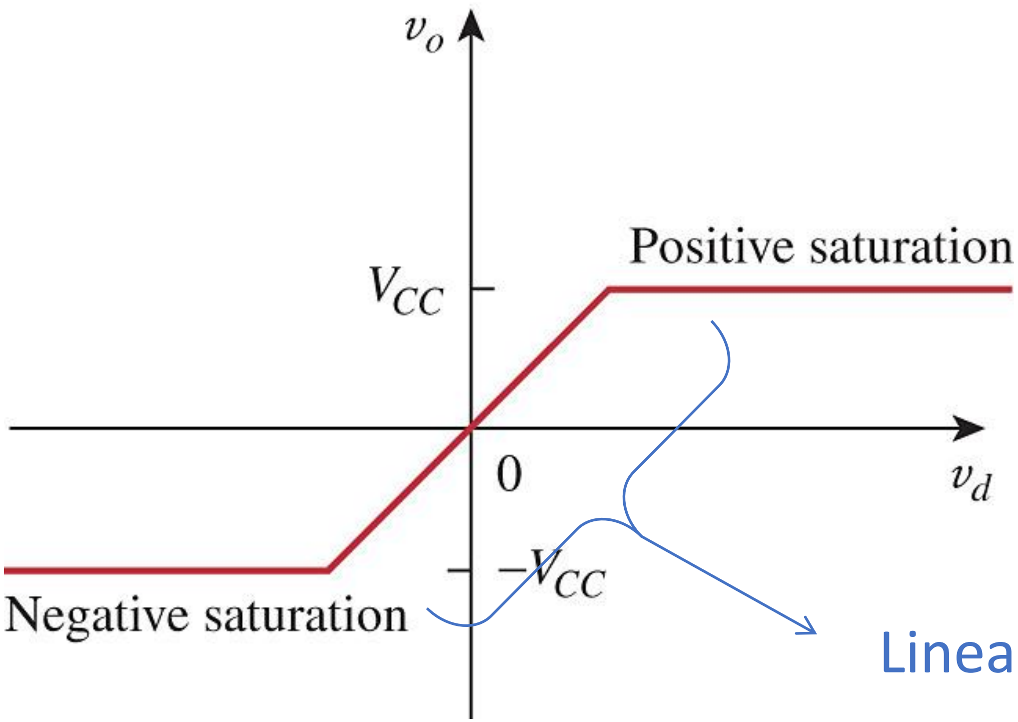
Supply voltage →

FEEDBACK:

- Negative feedback: output fed back to inverting input terminal
- Ratio of output voltage to input voltage = closed-loop gain



Practical limitations



$v_o = Vcc$

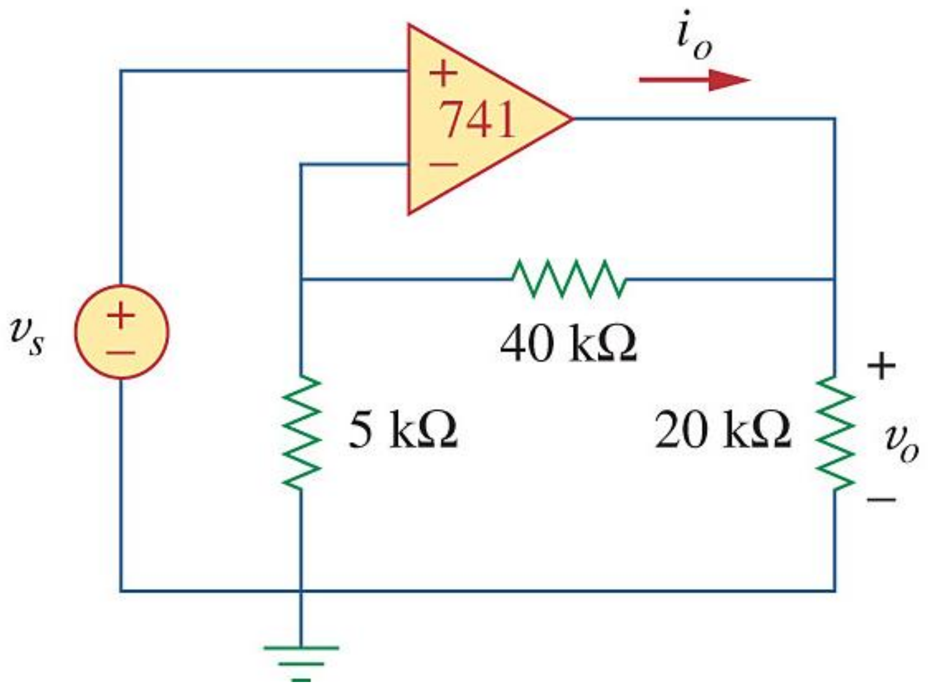
Linear region

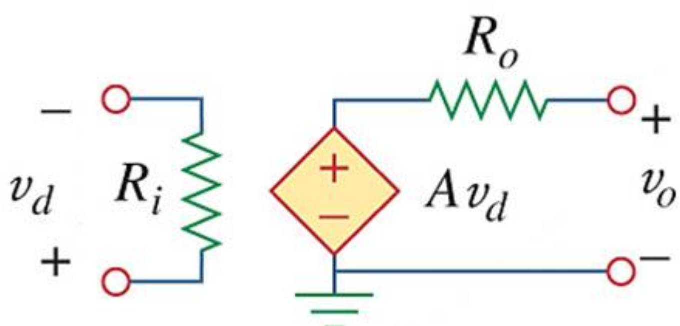
$-Vcc \leq v_o = Av_d \leq Vcc$

$v_o = -Vcc$

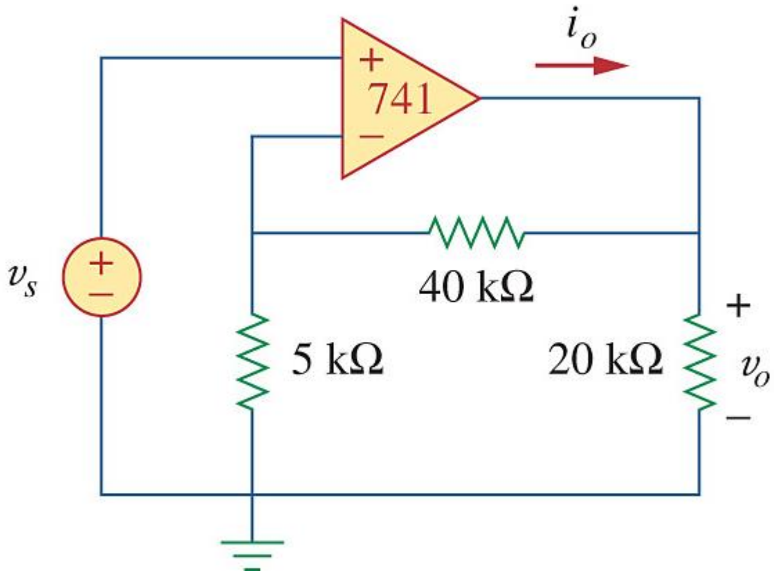
Example

Use a 741 Op Amp with $A = 2\times10^{5}$, R_i = 2 MΩ, R_o = 50 Ω.
Determine the closed-loop gain v_o/v_s

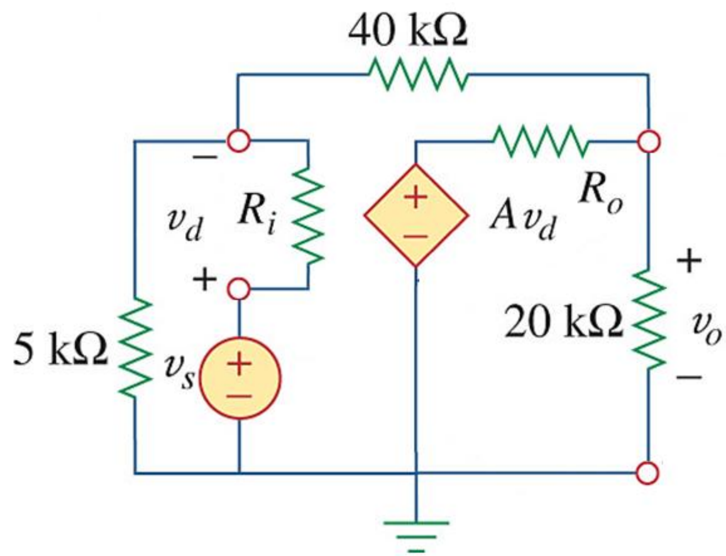




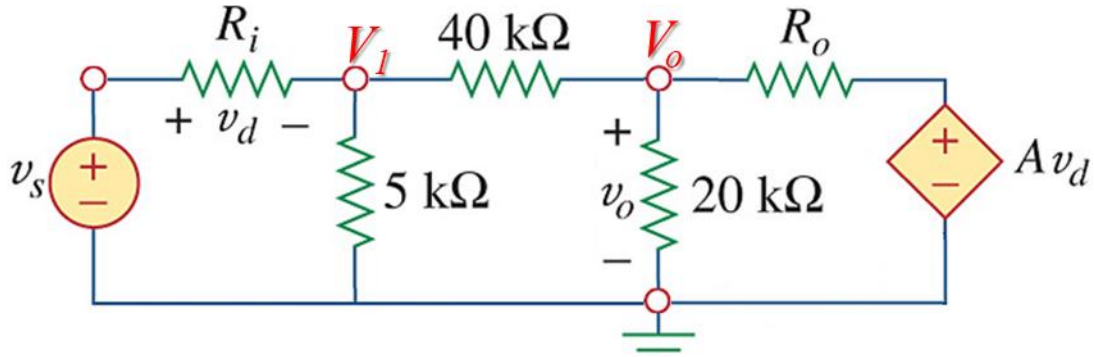
Example



i_o
741
v_s
40 kΩ
5 kΩ
20 kΩ
v_o



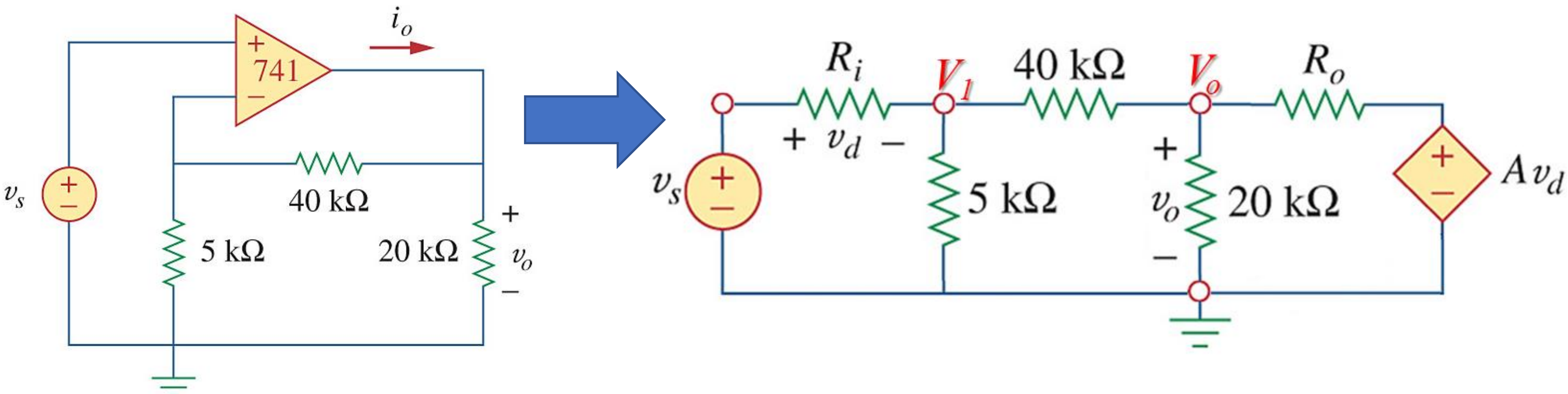
40 kΩ
v_d
R_i
Av_d
R_o
5 kΩ
v_s
20 kΩ
v_o



R_i
V_1
40 kΩ
V_o
R_o
v_d
v_s
5 kΩ
v_o
20 kΩ
Av_d

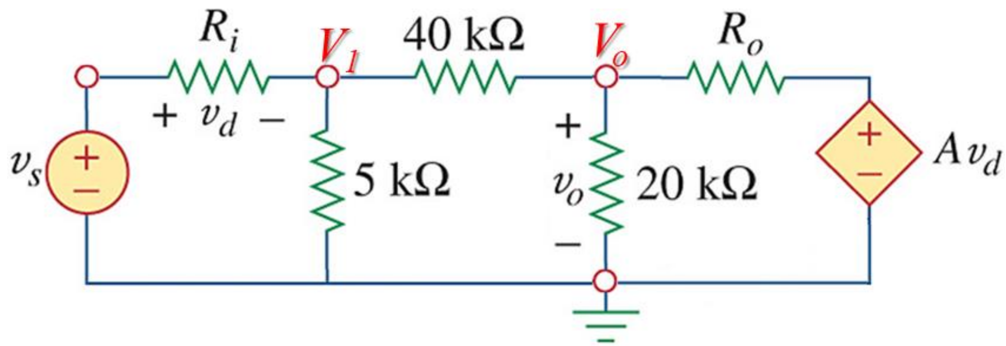
Example

Use a 741 Op Amp with $A = 2\times10^5$, $R_i = 2\ \mathrm{M}\Omega$, $R_o = 50\ \Omega$.
Determine the closed-loop gain v_o/v_s



Example

Use a 741 Op Amp with A = 2×10^5, R_i = 2 MΩ, R_o = 50 Ω. Determine the closed-loop gain v_o/v_s



KCL @ Node V_1

$$\frac{V_1 - V_s}{2\times10^6} + \frac{V_1}{5\times10^3} + \frac{V_1 - V_o}{40\times10^3} = 0$$

$$V_1 - V_s + 400V_1 + 50V_1 - 50V_o = 0$$

$$451V_1 - 50V_o = V_s \quad (1)$$

KCL @ Node V_o

$$\frac{V_o - V_1}{40\times10^3} + \frac{V_o}{20\times10^3} + \frac{V_o - Av_d}{50} = 0$$

$$V_o - V_1 + 2V_o + 800[V_o - A(V_s - V_1)] = 0$$

$$160\times10^6 V_1 + 803V_o = 160\times10^6 V_s$$

$$451V_1 + 2.263\times10^{-3}V_o = 451V_s \quad (2)$$

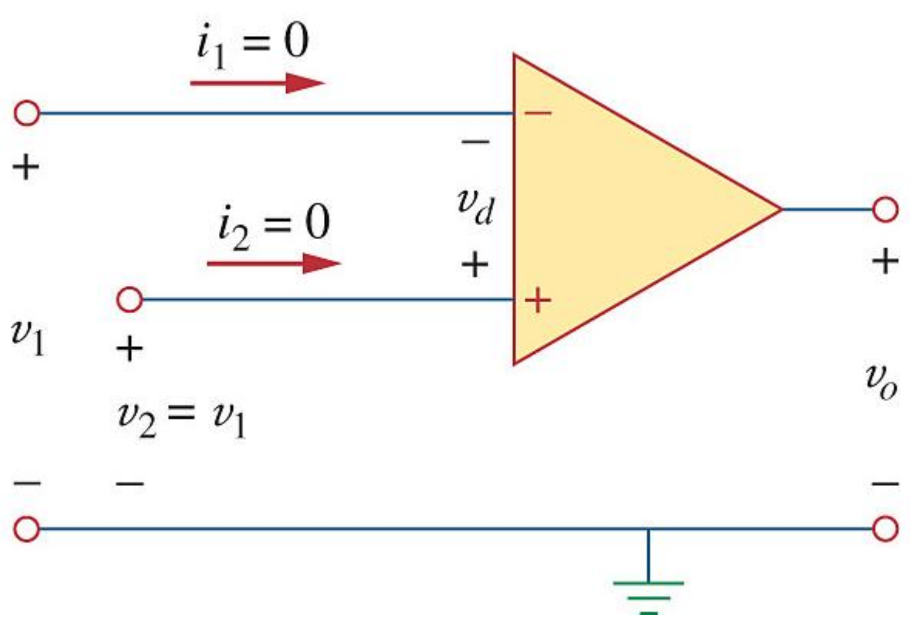
(1) – (2)

$$\boxed{\begin{gathered}-50.00226\, vo = -450V_s\\ \frac{v_o}{v_s} = \frac{-450}{-50.00226} = 8.9996\end{gathered}}$$

Ideal Op Amps

An ideal op amp has the following characteristics:

1. Infinite open-loop gain, $A \approx \infty$
2. Infinite input resistance, $R_i \approx \infty$
3. Zero output resistance, $R_o \approx 0$



Since $R_i = \infty$:

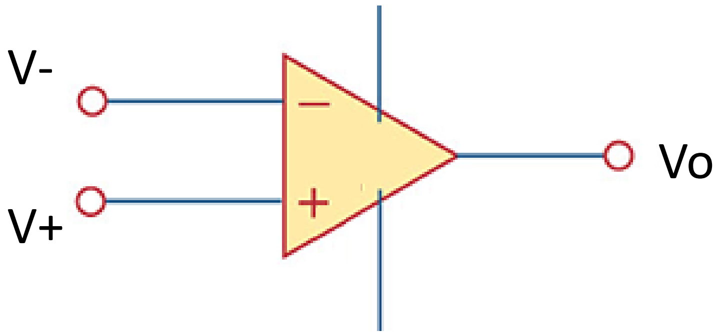
$$\boxed{i_1 = i_2 = 0}$$

$$\therefore v_d = v_2 - v_1 = 0$$

$$\boxed{\therefore\ v_2 = v_1}$$

Ideal vs non-ideal op-amps

IDEAL:

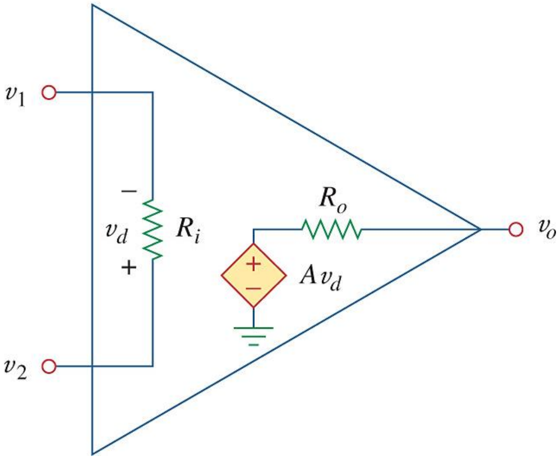


A	→	∞	V/V
Ri	→	∞	Ω
Ro	→	0	Ω

$i_1 = i_2 = 0$

$v_2 = v_1$

NON- IDEAL:

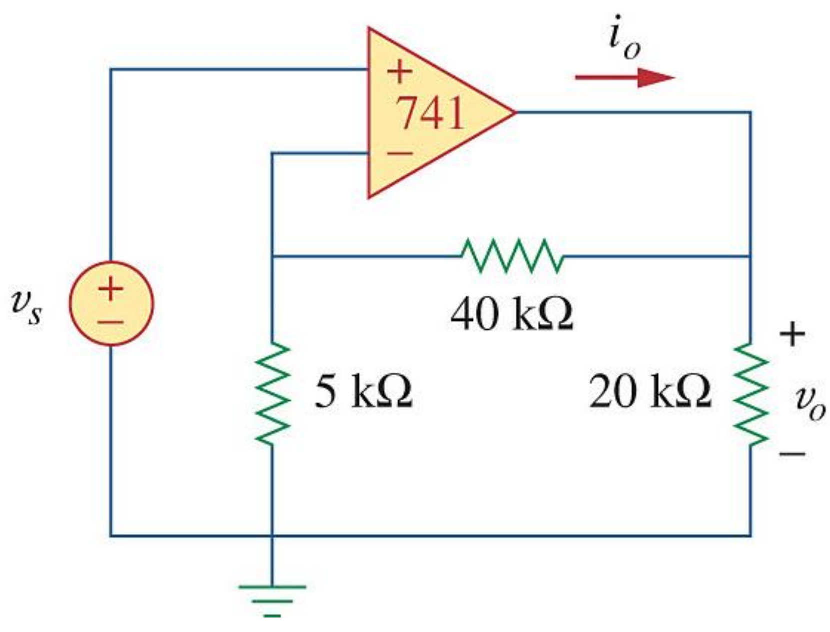


A	→	$10^5 – 10^8$	V/V
Ri	→	$10^5 – 10^{13}$	Ω
Ro	→	10 – 100	Ω

Example using ideal op-amp

Assume v_s = 1 V and determine v_o and i_o.

Remember:
$i_1 = i_2 = 0$
$v_+ = v-$



<u>KCL @ Node v- to get $\boldsymbol{V_o}$:</u>

$$\frac{1}{5 \times 10^3} + \frac{1 - V_o}{40 \times 10^3} = 0$$

$$8 + 1 - V_o = 0$$

$$V_o = 9\ \mathrm{V}$$

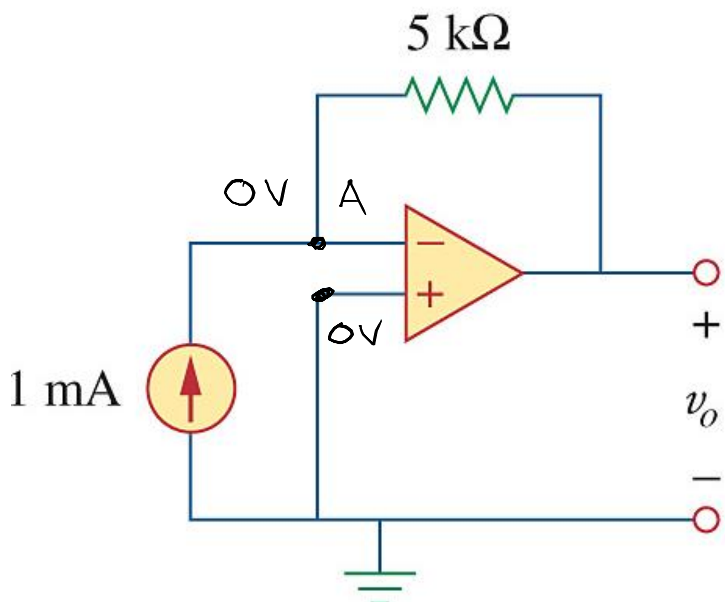
<u>KCL @ Node V_0 to get $\boldsymbol{i_o}$:</u>

$$i_o = \frac{9 - 1}{40 \times 10^3} + \frac{9}{20 \times 10^3}$$

$$i_o = 650\ \mu\mathrm{A}$$

Examples

Determine v_o



Node A: (KCL):

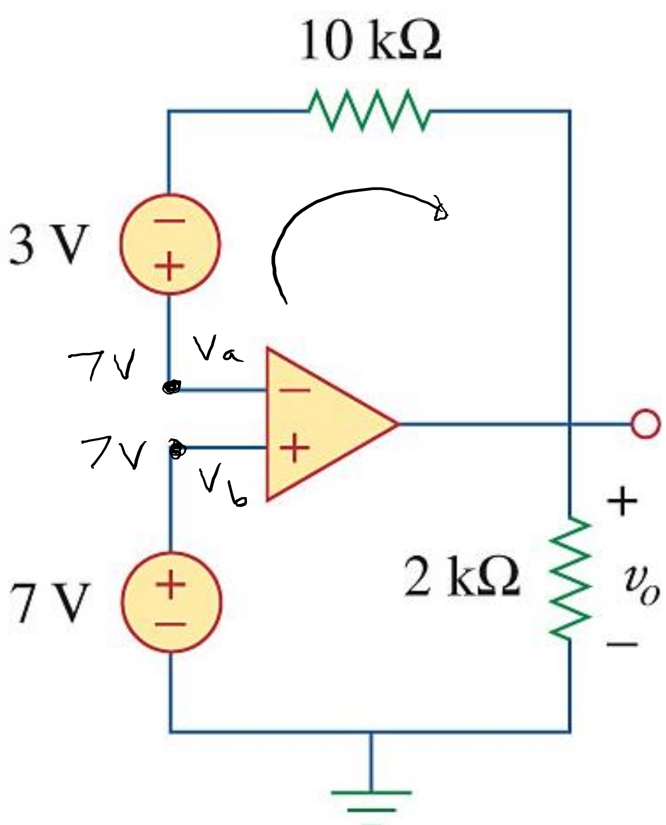
$$\frac{0 - V_o}{5k\Omega} + -1mA = 0$$

$$\therefore \frac{V_o}{5k\Omega} = -1mA$$

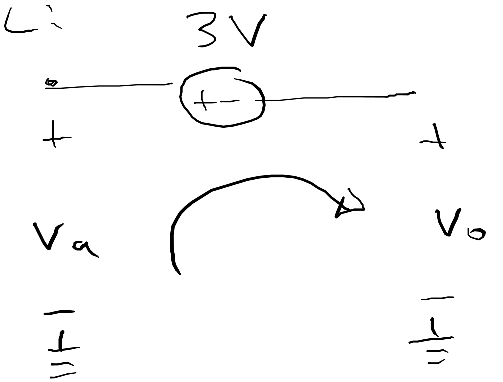
$$V_o = -5V$$

Examples

Determine v_o



KVL:



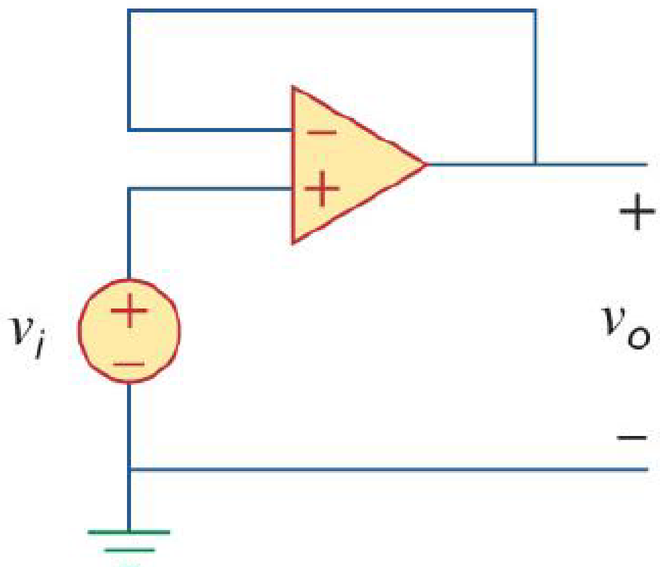
$$-V_a + 3 + V_o = 0$$

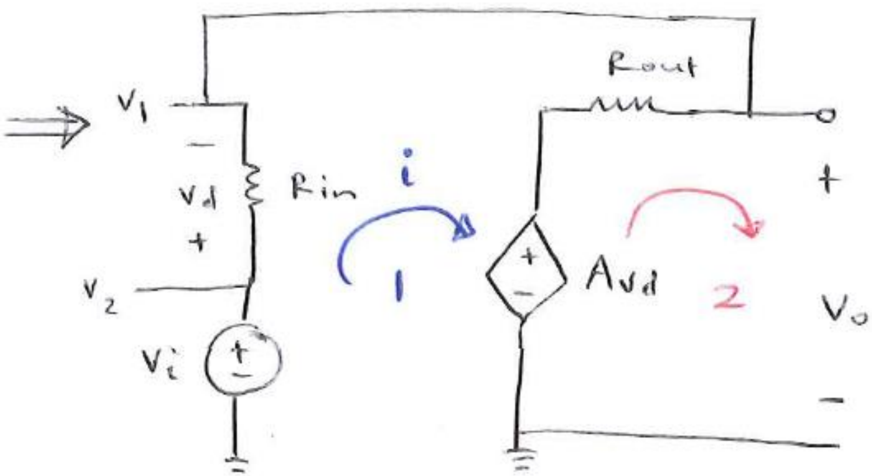
$$-7 + 3 + V_o = 0$$

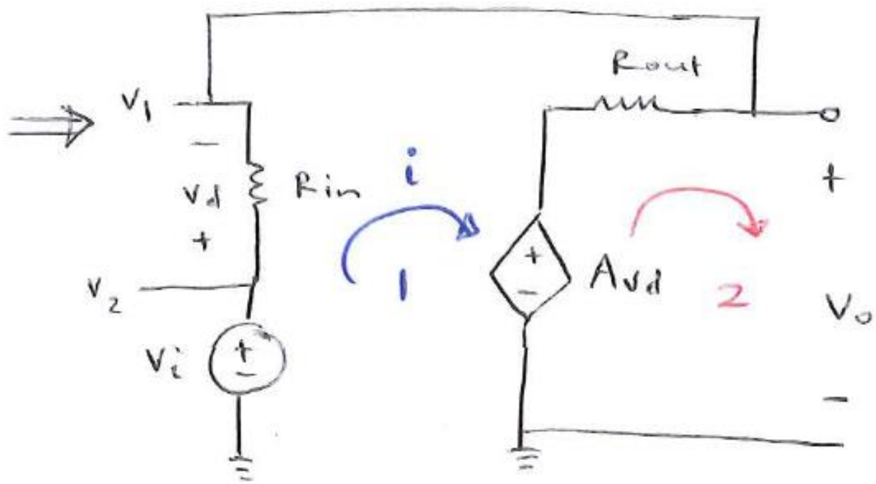
$$\therefore V_o = 4V$$

Example 1: Non-ideal op amp

- The op amp has an open-loop gain of 100 000, an input resistance of 10 kΩ, and an output resistance of 100 Ω. Find the voltage gain $\frac{v_o}{v_i}$.







KVL loop 1: $-V_i + V_d + R_{out}\, i + A v_d = 0$

and $V_d = R_{in}\, i$

$$\therefore \quad -v_i + i\,(R_{in} + R_{out}) + A\,(R_{in}\, i) = 0$$

$$\therefore \quad v_i = i\,(R_{in} + R_{out} + A R_{in})$$

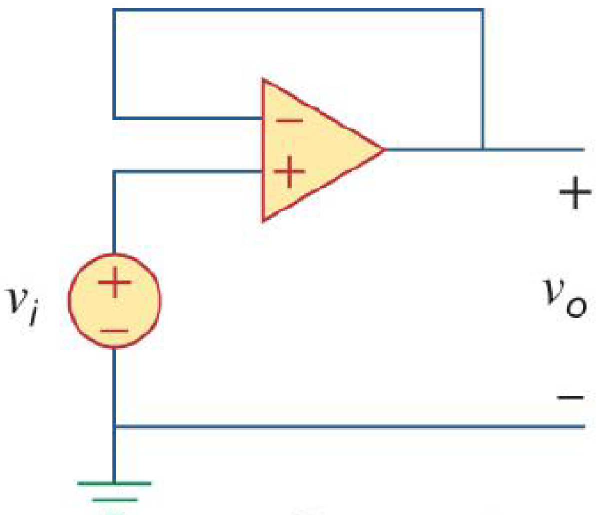
$$i = \frac{v_i}{R_{in}(1+A) + R_{out}} \quad (1)$$

KVL loop 2: $-A v_d + R_{out}(-i) + V_o = 0$

$$\therefore \quad V_o = A R_{in}\, i + R_{out}\, i = i\,(A R_{in} + R_{out}) \quad (2)$$

Example 1: Non-ideal op amp

- The op amp has an open-loop gain of 100 000, an input resistance of 10 kΩ, and an output resistance of 100 Ω. Find the voltage gain $\frac{v_o}{v_i}$.



① into ②:

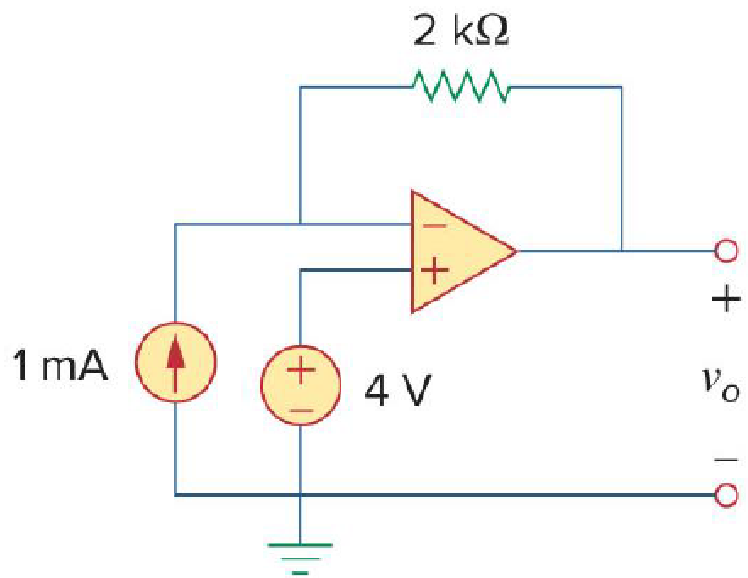
$$V_o = \frac{V_i\,(AR_{in} + R_{out})}{R_{in}(1+A) + R_{out}}$$

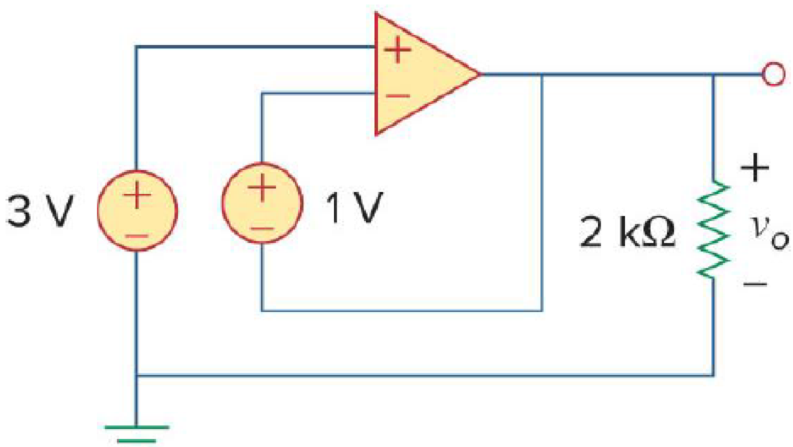
$$\therefore \frac{V_o}{V_i} = \frac{((100\times10^3 \cdot 10\times10^3) + 100)}{(10\times10^3(1 + 100\times10^3) + 100)} = \frac{1\times10^9}{1.00001\times10^9}$$

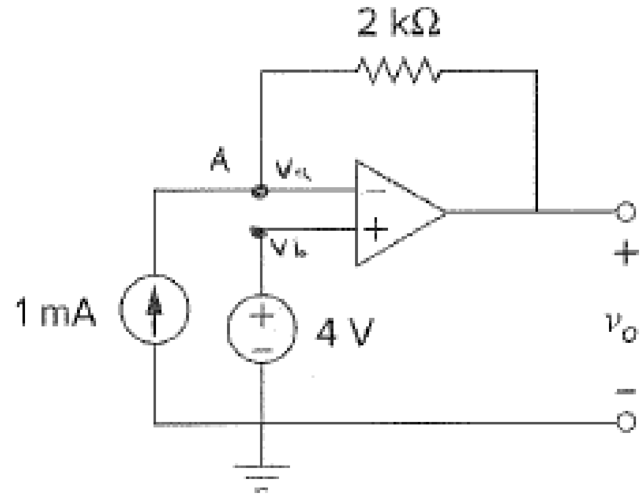
$$\frac{V_o}{V_i} = 0.999$$

Example 2: Ideal op amp

- Assume ideal op amps. Determine v_0 for both circuits.







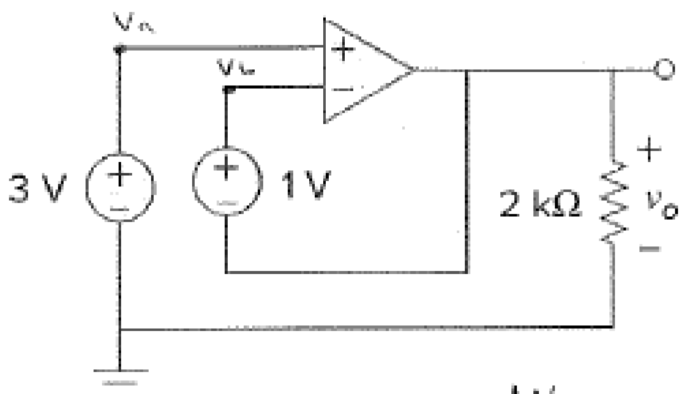
$V_a = V_b = 4V$

At node A: V_a

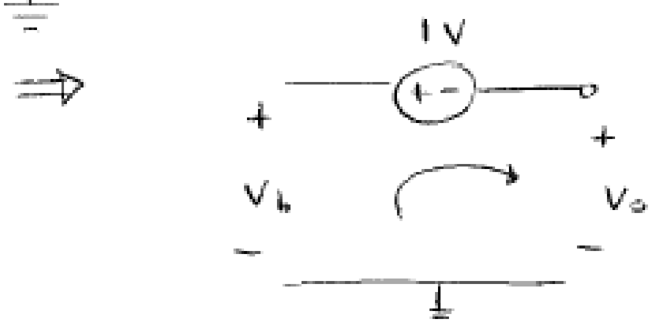
$$1mA = \frac{4 - V_o}{2k\Omega}$$

$$\therefore \; -V_o = (2k\Omega)(1mA) - 4$$

$$V_o = 4 - 2 = 2V$$



$V_a = V_b = 3V$



$$-V_b + 1 + V_o = 0$$

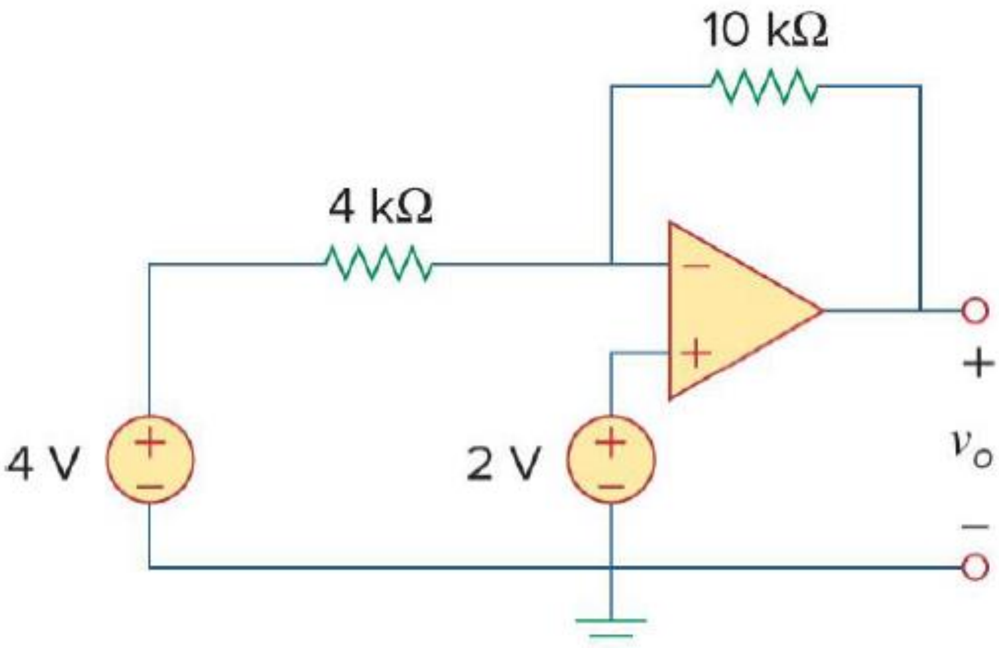
$$V_o = V_b - 1$$

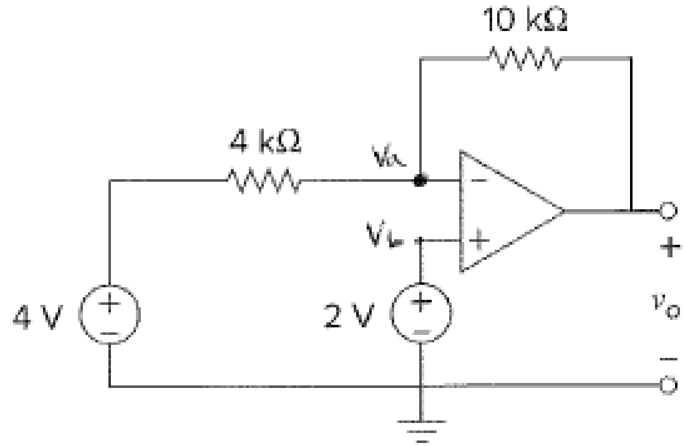
$$V_o = 3 - 1$$

$$V_o = 2V$$

Example 3: Ideal op amps

- Assume an ideal op amp. Determine v_0.





$V_a = V_b = 2\,V$

At node V_a:

$$\frac{V_a - 4}{4k} + \frac{V_a - V_o}{10k} = 0$$

$(\times 20k)$: $5V_a - 20 + 2V_a - 2V_o = 0$

and $V_a = 2\,V$:

$\therefore\ 7(2) - 20 - 2V_o = 0$

$\therefore\ 2V_o = -6$

$V_o = -3\,V$